Wave propagation through periodic arrays of freely floating rectangular floes: Supplementary Material.

A Preprint

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Abstract

Using the definitions and notation introduced in the journal submission, we outline details of the calculations carried out for surge and pitch motions and the associated hydrodynamic forces that depend on these motions.

1 Solution in surge

A general expression for $\phi^{(s)}(x,y)$ in the fluid region $V^+ := \{0 < x < \ell, -\hat{\rho}d < z < 0\}$ satisfying both (2.1) and (2.12) is given by

$$\phi^{(s)}(x,z) = \varphi^{(s)}(x,z) + \sum_{m=0}^{\infty} a_m Z_m(z) \cos \alpha_m x \tag{S.1.1}$$

where $\alpha_m = m\pi/\ell$ and

$$Z_0(z) = 1 + Kz, \quad Z_m(z) = \cosh \alpha_m z + (K/\alpha_m) \sinh \alpha_m z \tag{S.1.2}$$

are chosen to satisfy (2.2). In (S.1.1), we have defined the particular solution satisfying (2.1), (2.2) and (2.11) and the boundary condition $\varphi_z^{(s)}(x, -\hat{\rho}d) = 0$

$$\begin{aligned} \varphi^{(s)}(x,z) \quad = \quad & \frac{1}{2\ell}\left(x^2 - (\ell - x)^2 \mathrm{e}^{-\mathrm{i}kL}\right) - \frac{1}{2\ell}\left(1 - \mathrm{e}^{-\mathrm{i}kL}\right)\left(z^2 + 2\hat{\rho}dz\right) \\ & - \left(\frac{\hat{\rho}d}{K\ell} + \frac{\ell}{6}\right)\left(1 - \mathrm{e}^{-\mathrm{i}kL}\right) + \sum_{m=1}^{\infty} \frac{2K\left((-1)^m - \mathrm{e}^{-\mathrm{i}kL}\right)\cos(\alpha_m x)\cosh\left[\alpha_m (z + \hat{\rho}d)\right]}{\ell {\alpha_m}^2 \left(\alpha_m \sinh(\alpha_m \hat{\rho}d) - K\cosh(\alpha_m \hat{\rho}d)\right)}. \end{aligned} \tag{S.1.3}$$

Over $0 < x < \ell$ we let $W^{(s)}(x) = \phi_z^{(s)}(x, -\hat{\rho}d)$ to represent the unknown vertical component of the fluid velocity across the interface between the two rectangular fluid sub-domains V^+ and V^-, above and below the level $z = -\hat{\rho}d$.

Using the orthogonality of the functions $\cos \alpha_m x$ (see (3.3)) then gives

$$\left(K\cosh(\alpha_m \hat{\rho}d) - \alpha_m \sinh(\alpha_m \hat{\rho}d)\right) a_m = \frac{2}{\ell}\int_0^{\ell} W^{(s)}(x) \cos \alpha_m x \, \mathrm{d}x \tag{S.1.4}$$

for $m = 0, 1, \ldots$.

In $z < -\hat{\rho}d$, the general solution satisfying (2.1), (2.4) and (2.11) can be written

$$\phi^{(s)}(x,z) = \sum_{m=-\infty}^{\infty} b_m \mathrm{e}^{\mathrm{i}\beta_m x} \mathrm{e}^{\beta_m (z + \hat{\rho}d)} \tag{S.1.5}$$

where $\beta_m = k + 2\pi m/L$. Using (3.7) now gives us

$$b_m = \frac{1}{\beta_m L}\int_0^{\ell} W^{(s)}(x) \mathrm{e}^{-\mathrm{i}\beta_m x} \, \mathrm{d}x \tag{S.1.6}$$

for all m. To complete the solution we match $\phi^{(s)}$ from (S.1.1) and (S.1.5) across $z = -\hat{\rho}d$, $0 < x < \ell$ and substitute expressions for a_m and b_m from (S.1.3), (S.1.6), a process which results in the integral equation

$$(\mathcal{T}W^{(s)})(x) \equiv \frac{1}{\ell}\int_0^\ell W^{(s)}(x')T(x,x')\,\mathrm{d}x' = G^{(s)}(x), \quad 0 < x < \ell \tag{S.1.7}$$

for the unknown function $W^{(s)}(x)$ where $T(x,x')$ is defined by (3.18) and we have defined

$$\begin{aligned} G^{(s)}(x) = (1/\ell)\varphi^{(s)}(x,-\hat{\rho}d) &= \frac{x^2}{2\ell^2} - \frac{(\ell-x)^2}{2\ell^2}\mathrm{e}^{-\mathrm{i}kL} + \left(\frac{d^2}{\ell^2}\left(\frac{\hat{\rho}^2}{2} - \frac{\hat{\rho}}{Kd}\right) - \frac{1}{6}\right)(1-\mathrm{e}^{-\mathrm{i}kL}) \\ &\quad - \frac{2}{\ell^2}\sum_{m=1}^{\infty}\frac{((-1)^m - \mathrm{e}^{-\mathrm{i}kL})\cos\alpha_m x}{\alpha_m^2(\cosh(\alpha_m\hat{\rho}d) - (\alpha_m/K)\sinh(\alpha_m\hat{\rho}d))} \end{aligned} \tag{S.1.8}$$

which has been determined from (S.1.3).

In order to use the solution $W^{(s)}(x)$ of (S.1.7) to calculate the surge-induced surge force $F^{(s,s)}$ directly, we use Green's identity with the functions $\phi^{(s)}(x,z)$ and $\overline{\varphi^{(s)}}(x,z)$ in the region V^+ and this gives us

$$\begin{aligned} 0 &= \int_{\partial V^+}\frac{\partial\phi^{(s)}}{\partial n}\overline{\varphi^{(s)}} - \frac{\partial\overline{\varphi^{(s)}}}{\partial n}\phi^{(s)}\,\mathrm{d}s \\ &= -\int_0^\ell\left(\frac{\partial\phi^{(s)}}{\partial z}\overline{\varphi^{(s)}} - \frac{\partial\overline{\varphi^{(s)}}}{\partial z}\phi^{(s)}\right)\bigg|_{z=-\hat{\rho}d}\mathrm{d}x + \int_0^\ell\left(\frac{\partial\phi^{(s)}}{\partial z}\overline{\varphi^{(s)}} - \frac{\partial\overline{\varphi^{(s)}}}{\partial z}\phi^{(s)}\right)\bigg|_{z=0}\mathrm{d}x \\ &\quad + \int_{-\hat{\rho}d}^0\left(\frac{\partial\phi^{(s)}}{\partial x}\overline{\varphi^{(s)}} - \frac{\partial\overline{\varphi^{(s)}}}{\partial x}\phi^{(s)}\right)\bigg|_{x=\ell}\mathrm{d}z - \int_{-\hat{\rho}d}^0\left(\frac{\partial\phi^{(s)}}{\partial x}\overline{\varphi^{(s)}} - \frac{\partial\overline{\varphi^{(s)}}}{\partial x}\phi^{(s)}\right)\bigg|_{x=0}\mathrm{d}z \\ &= -\int_0^\ell W^{(s)}(x)\overline{\varphi^{(s)}}(x,-\hat{\rho}d)\,\mathrm{d}x + \int_{-\hat{\rho}d}^0\overline{\varphi^{(s)}}(\ell,z) - \phi^{(s)}(\ell,z)\,\mathrm{d}z - \int_{-\hat{\rho}d}^0\overline{\varphi^{(s)}}(0,z)\mathrm{e}^{-\mathrm{i}kL} - \phi^{(s)}(0,z)\mathrm{e}^{\mathrm{i}kL}\,\mathrm{d}z \end{aligned} \tag{S.1.9}$$

using the conditions satisfied by the two functions on the four boundaries and rearranging gives

$$\int_{-\hat{\rho}d}^0\phi^{(s)}(\ell,z) - \mathrm{e}^{\mathrm{i}kL}\phi^{(s)}(0,z)\,\mathrm{d}z = \int_{-\hat{\rho}d}^0\overline{\varphi^{(s)}}(\ell,z) - \overline{\varphi^{(s)}}(0,z)\mathrm{e}^{-\mathrm{i}kL}\,\mathrm{d}z - \int_0^\ell W^{(s)}(x)\overline{\varphi^{(s)}}(x,-\hat{\rho}d)\,\mathrm{d}x. \tag{S.1.10}$$

From the definition (2.13) we have

$$\begin{aligned} F^{(s,s)} &= \frac{1}{d(L-\ell)}\int_{-\hat{\rho}d}^0\phi^{(s)}(\ell,z) - \phi^{(s)}(0,z)\mathrm{e}^{\mathrm{i}kL}\,\mathrm{d}z \\ &= -\frac{\ell^2}{d(L-\ell)}A^{(s,s)} + C^{(s,s)} \end{aligned} \tag{S.1.11}$$

after substituting from (S.1.10) and using the definition (S.1.7). For convenience we have defined

$$A^{(s,s)} = \frac{1}{\ell}\int_0^\ell W^{(s)}(x)\overline{G^{(s)}}(x)\,\mathrm{d}x \tag{S.1.12}$$

and

$$C^{(s,s)} = \frac{1}{d(L-\ell)}\int_{-\hat{\rho}d}^0\overline{\varphi^{(s)}}(\ell,z) - \overline{\varphi^{(s)}}(0,z)\mathrm{e}^{-\mathrm{i}kL}\,\mathrm{d}x \tag{S.1.13}$$

in the above. Using the definition (S.1.3), we can calculate

$$C^{(s,s)} = \frac{\ell\hat{\rho}}{L-\ell} - \frac{4\hat{\rho}d^2\sin^2(kL/2))}{\ell(L-\ell)}\left(\frac{\ell^2}{6d^2} + \frac{\hat{\rho}}{Kd} - \frac{\hat{\rho}^2}{3}\right) + \frac{4}{\ell(L-\ell)d}\sum_{m=1}^{\infty}\frac{(1-(-1)^m\cos kL)}{\alpha_m^3((\alpha_m/K) - \coth(\alpha_m\hat{\rho}d))}. \tag{S.1.14}$$

1.1 Numerical approximation

Following the example of heave in the main paper, we write

$$W^{(s)}(x) \approx \sum_{n=0}^{N}A_n^{(s)}u_n(x) \tag{S.1.15}$$

for unknowns ${A_n}^{(s)}$ and with $u_n(x)$ defined by (3.30). After substituting into (S.1.7), multiplying by $\overline{u_m(x)}$ and integrating over $0 < x < \ell$ one obtains

$$\sum_{n=0}^{N} A_n^{(s)} T_{mn} = E_m^{(s)} \tag{S.1.16}$$

where T_{mn} is given by (3.28) whilst

$$E_m^{(s)} = \int_0^{\ell} G^{(s)}(x)\overline{u(x)}\,\mathrm{d}x. \tag{S.1.17}$$

From the definition (S.1.3) we find

$$E_m^{(s)} = \mathcal{K}_m + \left(-\frac{\hat{\rho}^2 d^2}{2\ell^2} + \frac{\hat{\rho} d}{K\ell^2} + \frac{1}{6}\right)\left(1 - \mathrm{e}^{-\mathrm{i}kL}\right)\overline{\mathcal{I}_{m0}} + \sum_{n=1}^{\infty} \frac{2K\left((-1)^n - \mathrm{e}^{-\mathrm{i}kL}\right)\overline{\mathcal{I}_{mn}}}{{\alpha_n}^2\ell^2\left(\alpha_n \sinh(\alpha_n\hat{\rho}d) - K\cosh(\alpha_n\hat{\rho}d)\right)}, \tag{S.1.18}$$

where $\mathcal{I}_{mn}$ is defined by (3.31), (3.32) and we have had to make a new calculation $\mathcal{K}_m = \langle g, u_m\rangle$ where $g(x) = (x^2 - (\ell - x)^2\mathrm{e}^{-\mathrm{i}kL})/(2\ell^2)$ and find, using some standard integral results for products of polynomials and Gegenbauer polynomials (REF), that

$$\mathcal{K}_m = \frac{\mathrm{i}^m 2^{1/6}\Gamma(8/3)}{\sqrt{\pi}\Gamma\left(m + 10/3\right)\Gamma(3 - m)}\left((-1)^m - \mathrm{e}^{-\mathrm{i}kL}\right) \tag{S.1.19}$$

noting that, for $m \geq 3$, we have $\mathcal{K}_m = 0$.

Finally, it follows from using (S.1.15) in (S.1.12) that

$$A^{(s,s)} \approx \sum_{n=0}^{N} A_n^{(s)}\overline{E_n^{(s)}}. \tag{S.1.20}$$

1.2 Asymptotic results for small gaps

Letting $\ell/d = \epsilon \ll 1$ and assuming $|K\hat{\rho}d - 1| \gg \epsilon$ (away from resonance) we have as before, from (3.40),

$$T(x, x') \approx \epsilon^{-1}\tilde{T}, \qquad \text{where } \tilde{T} = \frac{K\hat{\rho}d - 1}{Kd} + \frac{1}{2}\epsilon\cot\left(\frac{kL}{2}\right). \tag{S.1.21}$$

We also have from (S.1.8) the leading order behaviour

$$G^{(s)}(x) \approx \frac{1}{\epsilon^2}\tilde{G}^{(s)}, \qquad \text{where } \tilde{G}^{(s)} = \left(\frac{\hat{\rho}^2}{2} - \frac{\hat{\rho}}{Kd}\right)\left(1 - \mathrm{e}^{-\mathrm{i}kL}\right). \tag{S.1.22}$$

Using (S.1.21) and (S.1.22) in (S.1.7) gives

$$\frac{1}{\ell}\int_0^{\ell} W^{(s)}(x')\,\mathrm{d}x' \approx \frac{1}{\epsilon}\frac{\tilde{G}^{(s)}}{\tilde{T}} \tag{S.1.23}$$

which when used in (S.1.13) gives

$$A^{(s,s)} \approx \frac{1}{\epsilon^3}\frac{\left|\tilde{G}^{(s)}\right|^2}{\tilde{T}}. \tag{S.1.24}$$

Also, the leading order behaviour in (S.1.14) gives

$$C^{(s,s)} \approx \frac{1}{\epsilon}\frac{4\hat{\rho}^2}{KL}\left(\frac{1}{3}K\hat{\rho}d - 1\right)\sin^2\left(\frac{kL}{2}\right). \tag{S.1.25}$$

Using (S.1.24) and (S.1.25) with (S.1.21) and (S.1.22) in (S.1.11) and simplifying shows us that

$$F^{(s,s)} \approx \frac{1}{\epsilon}\frac{4\hat{\rho}^2 d}{KL}\sin^2\left(\frac{kL}{2}\right)\left(\frac{1}{3}K\hat{\rho}d - 1 - \frac{(K\hat{\rho}d - 2)^2}{4(K\hat{\rho}d - 1) + 2\epsilon Kd\cot(kL/2)}\right) \tag{S.1.26}$$

is the leading order behaviour of the surge-induced surge force for $\epsilon \ll 1$ away from resonance.

2 Combined heave-surge forces

We are now in a position to consider the heave-induced surge force and the surge-induced heave force (theory dictates that are complex conjugates).

We see from the definition (2.13) that

$$F^{(h,s)} = \frac{1}{d(L-\ell)} \int_{-\hat{\rho}d}^{0} \phi^{(h)}(\ell, z)\, \mathrm{d}z - \frac{\mathrm{e}^{\mathrm{i}kL}}{d(L-\ell)} \int_{-\hat{\rho}d}^{0} \phi^{(h)}(0, z)\, \mathrm{d}z \tag{S.2.1}$$

and

$$F^{(s,h)} = \frac{1}{d(L-\ell)} \int_{\ell}^{L} \phi^{(s)}(x, -\hat{\rho}d)\, \mathrm{d}x. \tag{S.2.2}$$

Applying Green's identity to $\phi^{(h)}(x,z)$ and $\overline{\varphi^{(s)}}(x,z)$ in the region V^+ we get, using all of the equations and boundary conditions defining those functions

$$\int_{-\hat{\rho}d}^{0} \phi^{(h)}(\ell, z)\, \mathrm{d}z - \mathrm{e}^{\mathrm{i}kL} \int_{-\hat{\rho}d}^{0} \phi^{(h)}(0, z)\, \mathrm{d}z = -\ell \int_{0}^{\ell} W^{(h)}(x)\overline{G^{(s)}(x)}\, \mathrm{d}x. \tag{S.2.3}$$

Instead, applying Green's identity to $\phi^{(s)}(x,z)$ and $\overline{\varphi^{(h)}}(x,z)$ in the region V^- and again using all of the equations and boundary conditions defining those functions

$$\int_{\ell}^{L} \phi^{(s)}(x, -\hat{\rho}d)\, \mathrm{d}x = -\ell \int_{0}^{\ell} W^{(s)}(x)\overline{G^{(h)}}(x)\, \mathrm{d}x. \tag{S.2.4}$$

Combining (S.2.3), (S.2.4) with (S.2.1), (S.2.2) results in

$$F^{(h,s)} = -\frac{\ell^2}{d(L-\ell)} A^{(h,s)}, \qquad \text{where } A^{(h,s)} = \frac{1}{\ell} \int_{0}^{\ell} W^{(h)}(x)\overline{G^{(s)}(x)}\, \mathrm{d}x \tag{S.2.5}$$

and

$$F^{(s,h)} = -\frac{\ell^2}{d(L-\ell)} A^{(s,h)}, \qquad \text{where } A^{(s,h)} = \int_{0}^{\ell} W^{(s)}(x)\overline{G^{(h)}(x)}\, \mathrm{d}x. \tag{S.2.6}$$

2.1 Numerical approximation

The numerical approximations to these quantities above are obtained from

$$A^{(h,s)} \approx \sum_{n=0}^{N} A_n^{(h)} \overline{E_n^{(s)}} \tag{S.2.7}$$

and

$$A^{(s,h)} \approx \sum_{n=0}^{N} A_n^{(s)} \overline{E_n^{(h)}}. \tag{S.2.8}$$

Use of the integral equations, $(\mathcal{T}W^{(s,h)})(x) = G^{(s,h)}(x)$, satisfied by $W^{(s,h)}(x)$ gives

$$A^{(h,s)} = \langle W^{(h)}, G^{(s)} \rangle = \langle W^{(h)}, \mathcal{T}W^{(s)} \rangle = \langle \mathcal{T}W^{(h)}, W^{(s)} \rangle = \langle G^{(h)}, W^{(s)} \rangle = \overline{A^{(s,h)}}. \tag{S.2.9}$$

This means $F^{(h,s)} = \overline{F^{(s,h)}}$ confirming the identity (2.17) in the main paper and thus we only need consider $F^{(h,s)}$.

2.2 Asymptotic analysis for small gaps

Using (3.37), (S.1.21) and (S.1.22) we can immediately write

$$A^{(h,s)} \approx \frac{1}{\epsilon^2} \frac{\tilde{G}^{(h)}\overline{\tilde{G}^{(s)}}}{\tilde{T}}, \qquad \text{and} \qquad A^{(s,h)} \approx \frac{1}{\epsilon^2} \frac{\tilde{G}^{(s)}\overline{\tilde{G}^{(h)}}}{\tilde{T}} \tag{S.2.10}$$

and so, by (S.2.5), (S.2.6) the leading order approximation of the surge/heave forces for small gaps are defined by the expression

$$F^{(h,s)} = i\hat{\rho} \frac{\frac{1}{2}K\hat{\rho}d - 1}{K\hat{\rho}d - 1 + \frac{1}{2}\epsilon K d \cot\left(\frac{kL}{2}\right)}. \tag{S.2.11}$$

3 Solution in Pitch

This is the most complicated problem to solve because the forcing applies on all sides of the submerged rectangular floe and so particular solutions are required in both V^+ and V^-.

A general expression for the potential $\phi^{(p)}(x,z)$ in the fluid region $V^+ := \{0 < x < \ell, -\hat{\rho}d < z < 0\}$ satisfying (2.1), (2.2) and (2.12) can be written as

$$\phi^{(p)}(x,z) = \varphi^{(+p)}(x,z) + \sum_{m=0}^{\infty} a_m Z_m(z) \cos \alpha_m x \tag{S.3.1}$$

where $\alpha_m = m\pi/\ell$ and

$$Z_0(z) = 1 + Kz, \quad Z_m(z) = \cosh \alpha_m z + (K/\alpha_m) \sinh \alpha_m z \tag{S.3.2}$$

are defined as before. The function $\varphi^{(+p)}(x,z)$ satisfies The function $\varphi^{(+p)}(x,z)$ satisfies (2.1), (2.2) and the inhomogeneous condition (2.11) as well as the imposed condition $\psi_z^{(+p)}(x,-\hat{\rho}d) = 0$. After considerable effort, we find it can be written

$$\begin{aligned}\varphi^{(+p)}(x,z) \quad = \quad & \frac{1}{2\ell}\left(x^2 - (\ell - x)^2 \mathrm{e}^{-\mathrm{i}kL}\right)\frac{(z - z^c)}{L - \ell} - \frac{1}{6\ell}\left(1 - \mathrm{e}^{-\mathrm{i}kL}\right)\frac{((z - z^c)^3 - 3\left(z^c + \hat{\rho}d\right)^2 z + \ell^2(z - z^c))}{L - \ell} \\ & + \frac{(3\hat{\rho}d(2z^c + \hat{\rho}d) - Kz^{c\,3})}{6K(L - \ell)\ell}\left(1 - \mathrm{e}^{-\mathrm{i}kL}\right) \\ & - \frac{2}{\ell(L - \ell)}\sum_{m=1}^{\infty}\frac{((-1)^m - \mathrm{e}^{-\mathrm{i}kL})}{\alpha_m^3 \cosh(\alpha_m \hat{\rho}d)}\cos\left(\alpha_m x\right)\left(\sinh\left[\alpha_m z\right] + C_m \cosh\left[\alpha_m(z + \hat{\rho}d)\right]\right)\end{aligned} \tag{S.3.3}$$

where

$$C_m = \frac{\alpha_m(\mathrm{sech}(\alpha_m \hat{\rho}d)) - 1 - Kz^c)}{K - \alpha_m \tanh(\alpha_m \hat{\rho}d)}. \tag{S.3.4}$$

We are reminded that $z^c = (\frac{1}{2} - \hat{\rho})d$ is the z coordinate of the centre of mass. In deriving this expression we have, in turn, satisfied the inhomogenous boundary conditions (2.11) on $x = 0$, $x = \ell$, and then Laplace's equation, followed by the condition on $z = -\hat{\rho}d$ and finally the condition on $z = 0$.

After taking the z-derivative of (S.3.1), evaluated on $z = -\hat{\rho}d$, multiplying by $\cos \alpha_m x$ and integrating over $0 < x < \ell$ we find, for $m = 0$,

$$a_0 = \frac{1}{K\ell}\int_0^{\ell} W^{(p)}(x)\,\mathrm{d}x \tag{S.3.5}$$

where we have let $W^{(p)}(x) = \phi_z^{(p)}(x,-\hat{\rho}d)$ for $0 < x < \ell$. For $m \geq 1$,

$$\left(K\cosh\left(\alpha_m \hat{\rho}d\right) - \alpha_m \sinh\left(\alpha_m \hat{\rho}d\right)\right) a_m = \frac{2}{\ell}\int_0^{\ell} W^{(p)}(x)\cos \alpha_m x\,\mathrm{d}x. \tag{S.3.6}$$

In $z < -\hat{\rho}d$, we require a general solution to satisfy (2.1), (2.4) and (2.11) and thus write

$$\phi^{(p)}(x,z) = \varphi^{(-p)}(x,z) + \sum_{m=-\infty}^{\infty} b_m \mathrm{e}^{\mathrm{i}\beta_m x}\mathrm{e}^{\beta_m(z+\hat{\rho}d)} \tag{S.3.7}$$

where $\beta_m = k + 2\pi m/L$ as before. Here, we have defined $\varphi^{(-p)}(x,z)$ to be the harmonic function satisfying (2.4) and decaying into the depth in addition to the imposed condition

$$\varphi_z^{(-p)}(x,-\hat{\rho}d) = \begin{cases} -(x - x^c)/(L - \ell), & \ell < x < L \\ 0, \quad 0 < x < \ell \end{cases} \tag{S.3.8}$$

which accounts for the forcing on the underside of the floe in much the same way as $\varphi^{(+p)}(x,z)$ accounted for the forcing on the sides of the floe in V^+.

The particular solution can be determined by writing the separation series satisfying (2.1) and (2.4)

$$\varphi^{(-p)}(x,z) = \ell\sum_{m=-\infty}^{\infty}\frac{g_m^{(p)}\mathrm{e}^{\mathrm{i}\beta_m x}}{\beta_m L}\mathrm{e}^{\beta_m(z+\hat{\rho}d)} \tag{S.3.9}$$

where (S.3.8) tells us that

$$\ell g_m^{(p)} = -\frac{1}{(L-\ell)}\int_\ell^L (x-x^c)\mathrm{e}^{-\mathrm{i}\beta_m x}\,\mathrm{d}x = \frac{\mathrm{e}^{-\mathrm{i}\beta_m \ell}-\mathrm{e}^{-\mathrm{i}\beta_m L}}{{\beta_m}^2(L-\ell)} + \frac{1}{2}\frac{\mathrm{e}^{-\mathrm{i}\beta_m \ell}+\mathrm{e}^{-\mathrm{i}\beta_m L}}{\mathrm{i}\beta_m} \tag{S.3.10}$$

after using the orthogonality of the functions $\mathrm{e}^{\mathrm{i}\beta_m x}$ on $0 < x < L$ expressed by (3.10). As a result of taking the z-derivative of (S.3.7) on $z = -\hat{\rho}d$, we find

$$b_m = \frac{1}{\beta_m L}\int_0^\ell W^{(p)}(x)\mathrm{e}^{-\mathrm{i}\beta_m x}\,\mathrm{d}x \tag{S.3.11}$$

for all m. Next, applying continuity of $\phi^{(p)}$ across the common interface $z = -\hat{\rho}d$, $0 < x < \ell$ between the two regions and substituting for a_m and b_m from (S.3.6), (S.3.11) results in the integral equation

$$(\mathcal{T}W^{(p)})(x) \equiv \frac{1}{\ell}\int_0^\ell W^{(p)}(x')T(x,x')\,\mathrm{d}x' = G^{(p)}(x), \quad 0 < x < \ell \tag{S.3.12}$$

where $T(x,x')$ is defined by (3.18) and we have defined the function

$$G^{(p)}(x) = (1/\ell)\left(\varphi^{(+p)}(x,-\hat{\rho}d) - \varphi^{(-p)}(x,-\hat{\rho}d)\right). \tag{S.3.13}$$

Using Green's identity with the functions two harmonic functions $\phi^{(p)}(x,z)$ and $\overline{\varphi^{(-p)}}(x,z)$ in the region V^- and using all of the conditions of these functions on the boundary of V^- gives us

$$\begin{aligned} 0 &= \int_{V^-}\frac{\partial\phi^{(p)}}{\partial n}\overline{\varphi^{(-p)}} - \frac{\partial\overline{\varphi^{(-p)}}}{\partial n}\phi^{(p)}\,\mathrm{d}s \\ &= \int_0^\ell\left(\frac{\partial\phi^{(p)}}{\partial z}\overline{\varphi^{(-p)}} - \frac{\partial\overline{\varphi^{(-p)}}}{\partial z}\phi^{(p)}\right)\Bigg|_{z=-\hat{\rho}d}\mathrm{d}x + \int_\ell^L\left(\frac{\partial\phi^{(p)}}{\partial z}\overline{\varphi^{(-p)}} - \frac{\partial\overline{\varphi^{(-p)}}}{\partial z}\phi^{(p)}\right)\Bigg|_{z=-\hat{\rho}d}\mathrm{d}x \\ &= \int_0^\ell W^{(h)}(x)\overline{\varphi^{(-p)}}(x,-\hat{\rho}d)\,\mathrm{d}x - \int_\ell^L\frac{x-x^c}{L-\ell}\left(\overline{\varphi^{(-p)}}(x,-\hat{\rho}d) - \phi(x,-\hat{\rho}d)\right)\mathrm{d}x \end{aligned} \tag{S.3.14}$$

from which we can clearly see

$$\int_\ell^L\frac{x-x^c}{L-\ell}\phi^{(p)}(x,-\hat{\rho}d)\,\mathrm{d}x = -\int_0^\ell W^{(p)}(x)\overline{\varphi^{(-p)}}(x,-\hat{\rho}d)\,\mathrm{d}x + \int_\ell^L\frac{x-x^c}{L-\ell}\overline{\varphi^{(-p)}}(x,-\hat{\rho}d)\,\mathrm{d}x. \tag{S.3.15}$$

Separately we use Green's identity with $\phi^{(p)}(x,z)$ and $\overline{\varphi^{(+p)}}(x,z)$ in V^+ which results, after some algebra, in the relation

$$\begin{aligned} \int_{-\hat{\rho}d}^0\frac{z-z^c}{L-\ell}\left(\phi^{(p)}(\ell,z) - \mathrm{e}^{\mathrm{i}kL}\phi^{(p)}(0,z)\right)\mathrm{d}z &= -\int_0^\ell W^{(p)}(x)\overline{\varphi^{(+p)}}(x,-\hat{\rho}d)\,\mathrm{d}x \\ &\quad + \int_{-\hat{\rho}d}^0\frac{z-z^c}{L-\ell}\left(\overline{\varphi^{(+p)}}(\ell,z) - \overline{\varphi^{(+p)}}(0,z)\mathrm{e}^{-\mathrm{i}kL}\right)\mathrm{d}z. \end{aligned} \tag{S.3.16}$$

Now, from the general formula (2.13) we can deduce that the pitch-induced pitch force is expressed in terms of the potential $\phi^{(p)}$ as

$$F^{(p,p)} = -\frac{1}{d(L-\ell)^2}\int_\ell^L\phi^{(p)}(x,-\hat{\rho}d)(x-x^c)\,\mathrm{d}x + \frac{1}{d(L-\ell)^2}\int_{-\hat{\rho}d}^0(z-z^c)\left(\phi^{(p)}(\ell,z) - \phi^{(p)}(0,z)\mathrm{e}^{\mathrm{i}kL}\right)\mathrm{d}z. \tag{S.3.17}$$

Using (S.3.15) and (S.3.16) in (S.3.17) along with the definition (S.3.13) gives

$$\begin{aligned} F^{(p,p)} &= -\frac{\ell}{d(L-\ell)}\int_0^\ell W^{(p)}(x)\overline{G^{(p)}(x)}\,\mathrm{d}x - \frac{1}{d(L-\ell)^2}\int_\ell^L(x-x^c)\overline{\varphi^{(-p)}}(x,-\hat{\rho}d)\,\mathrm{d}x \\ &\quad + \frac{1}{d(L-\ell)^2}\int_{-\hat{\rho}d}^0(z-z^c)\left(\overline{\varphi^{(+p)}}(\ell,z) - \overline{\varphi^{(+p)}}(0,z)\mathrm{e}^{-\mathrm{i}kL}\right)\mathrm{d}z. \end{aligned} \tag{S.3.18}$$

Thus, we have written $F^{(p,p)}$ in terms of the solution, $W^{(p)}(x)$, of (S.3.12) plus integrals of known functions. It helps to write

$$F^{(p,p)} = -\frac{\ell^2}{d(L-\ell)}A^{(p,p)} + C^{(p,p)} \tag{S.3.19}$$

where

$$A^{(p,p)} = \frac{1}{\ell}\int_0^{\ell} W^{(p)}(x)\overline{G^{(p)}(x)}\,\mathrm{d}x \tag{S.3.20}$$

and

$$C^{(p,p)} = C_1^{(p,p)} + C_2^{(p,p)} \tag{S.3.21}$$

with: (i)

$$C_1^{(p,p)} = -\frac{1}{d(L-\ell)^2}\int_{\ell}^{L}(x-x^c)\overline{\varphi^{(-p)}}(x,-\hat{\rho}d)\,\mathrm{d}x = \frac{\ell^2}{d(L-\ell)}\sum_{m=-\infty}^{\infty}\frac{|g_m^{(p)}|^2}{\beta_m L} = 0 \tag{S.3.22}$$

a result which is proved in the Appendix; and (ii)

$$C_2^{(p,p)} = \frac{1}{d(L-\ell)^2}\int_{-\hat{\rho}d}^{0}(z-z^c)\left(\overline{\varphi^{(+p)}}(\ell,z) - \overline{\varphi^{(+p)}}(0,z)\mathrm{e}^{-\mathrm{i}kL}\right)\,\mathrm{d}z. \tag{S.3.23}$$

A considerable amount of algebra allows us to determine

$$\begin{aligned} C^{(p,p)} \;=\; & \frac{\ell d^2\hat{\rho}}{(L-\ell)^3}\left(\frac{\hat{\rho}^2}{3}-\frac{\hat{\rho}}{2}+\frac{1}{4}\right) + \frac{d^4\sin^2(kL/2)}{\ell(L-\ell)^3}\left(\frac{\hat{\rho}^5}{5}-\frac{\hat{\rho}^4}{2}+\frac{\hat{\rho}^3}{3}-\frac{\hat{\rho}^2(1-\hat{\rho})^2}{Kd}-\frac{\ell^2}{d^2}\left(\frac{2\hat{\rho}^3}{9}-\frac{\hat{\rho}^2}{3}+\frac{\hat{\rho}}{6}\right)\right) \\ & -\frac{4}{d\ell(L-\ell)^3}\sum_{m=1}^{\infty}\frac{(1-(-1)^m\cos(kL))}{\alpha_m^5\cosh(\alpha_m\hat{\rho}d)}\left(\alpha_n\hat{\rho}d-(1+\tfrac{1}{2}C_m\alpha_m d)(1-\mathrm{e}^{-\alpha_m\hat{\rho}d})\right). \end{aligned} \tag{S.3.24}$$

3.1 Numerical approximation

We follow the same methods as before, and start by writing

$$W^{(p)}(x) \approx \sum_{n=0}^{N} A_n^{(p)} u_n(x) \tag{S.3.25}$$

which, when used in (S.3.11) under Galerkin's method, results in the expansion coefficients $A_n^{(p)}$ being determined by the linear system of equations

$$\sum_{n=0}^{N} A_n^{(p)} T_{mn} = E_m^{(p)}, \tag{S.3.26}$$

where

$$E_m^{(p)} = \frac{1}{\ell}\int_0^{\ell} G^{(p)}(x)\overline{u_m(x)}\,\mathrm{d}x. \tag{S.3.27}$$

Using the definitions of $G^{(p)}(x)$ and $u_m(x)$ from (S.3.13) and ((3.35)) we find, again after considerable algebra,

$$\begin{aligned} E_m^{(p)} \;=\; & -\sum_{n=-\infty}^{\infty}\frac{g_n^{(p)}}{\beta_n L}\overline{\mathcal{J}_{mn}} - \frac{d}{2(L-\ell)}\mathcal{K}_m + \frac{d^3(1-\mathrm{e}^{-\mathrm{i}kL})}{6\ell^2(L-\ell)}\left(\frac{3\hat{\rho}(1-\hat{\rho})}{Kd}-\frac{\ell^2}{2d^2}+\hat{\rho}^3-\frac{3\hat{\rho}^2}{2}\right)\overline{\mathcal{I}_{m0}} \\ & -\frac{2}{\ell^2(L-\ell)}\sum_{n=1}^{\infty}\frac{((-1)^n-\mathrm{e}^{-\mathrm{i}kL})}{\alpha_n^3\cosh(\alpha_n\hat{\rho}d)}\left(C_n-\sinh(\alpha_n\hat{\rho}d)\right)\mathcal{I}_{mn}. \end{aligned} \tag{S.3.28}$$

In the above $\mathcal{K}_m$ is defined by (S.1.19) and $\mathcal{J}_{mn}$ and $\mathcal{I}_{mn}$ are defined in the main paper.

Finally, from (S.3.26) in (S.3.28) we have

$$A^{(p,p)} \approx \sum_{n=0}^{N} A_n^{(p)}\overline{E_m^{(p)}}. \tag{S.3.29}$$

3.2 Asymptotic results for small gaps

First we note from the definition (S.3.13) that we can write

$$G^{(p)} \approx \frac{1}{\epsilon^2}\tilde{G}^{(p)} \tag{S.3.30}$$

to leading order where

$$\tilde{G}^{(p)} = \frac{\hat{\rho} d}{2L}(1 - \mathrm{e}^{-\mathrm{i}kL})\left(\frac{\hat{\rho}^2}{3} - \frac{\hat{\rho}}{2} + \frac{(1-\hat{\rho})}{Kd}\right). \tag{S.3.31}$$

Using $T(x, x') \approx \epsilon^{-1}\tilde{T}$ where $\tilde{T}$ is given by (3.35), the solution of (S.3.12) is approximated by

$$\frac{1}{\ell}\int_0^{\ell} W^{(p)}(x')\,\mathrm{d}x' \approx \frac{1}{\epsilon}\frac{\tilde{G}^{(p)}}{\tilde{T}}. \tag{S.3.32}$$

It follows from (S.3.20) that

$$A^{(p,p)} \approx \frac{1}{\epsilon^3}\frac{|\tilde{G}^{(p)}|^2}{\tilde{T}}. \tag{S.3.33}$$

Simultaneously, we can show that the leading order behaviour of $C^{(p,p)}$ defined in (S.3.24), is

$$C^{(p,p)} \approx \frac{1}{\epsilon}\frac{d^3}{L^3}\hat{\rho}^2 \sin^2(kL/2)\left(\frac{\hat{\rho}^3}{5} - \frac{\hat{\rho}^2}{2} + \frac{\hat{\rho}}{3} - \frac{(1-\hat{\rho})^2}{Kd}\right). \tag{S.3.34}$$

Bringing both results together in (S.3.19) gives

$$F^{(p,p)} = \frac{1}{\epsilon}\hat{\rho}^2\left(\frac{d}{L}\right)^3\left(\frac{\hat{\rho}^3}{5} - \frac{\hat{\rho}^2}{2} + \frac{\hat{\rho}}{3} - \frac{(1-\hat{\rho})^2}{Kd} - \frac{\left(\frac{\hat{\rho}^2}{3} - \frac{\hat{\rho}}{2} + \frac{1-\hat{\rho}}{Kd}\right)^2}{\frac{K\hat{\rho}d-1}{Kd} + \frac{1}{2}\epsilon\cot\left(\frac{kL}{2}\right)}\right)\sin^2\left(\frac{kL}{2}\right) \tag{S.3.35}$$

as the leading order approximation to the pitch-induced pitch force away from resonance.

4 Combined heave-pitch forces

From (2.13) we express the pitch-induced heave force as

$$F^{(p,h)} = \frac{1}{d(L-\ell)}\int_{\ell}^{L} \phi^{(p)}(x, -\hat{\rho}d)\,\mathrm{d}x \tag{S.4.1}$$

after using the conditions satisfied by the $\phi^{(h)}$ on $z = -\hat{\rho}d$, $x = 0, \ell$. In order to express this in terms of solutions to the integral equations formulated for each problem, we use $\phi^{(p)}$ and $\varphi^{(h)}$ over V^- in Green's identity to get

$$\begin{aligned} 0 &= \int_{\partial V^-} \frac{\partial \phi^{(p)}}{\partial n}\overline{\varphi^{(h)}} - \frac{\partial \overline{\varphi^{(h)}}}{\partial n}\phi^{(p)}\,\mathrm{d}s \\ &= \int_0^{\ell}\left(\frac{\partial \phi^{(p)}}{\partial z}\overline{\varphi^{(h)}} - \frac{\partial \overline{\varphi^{(h)}}}{\partial z}\phi^{(p)}\right)\Bigg|_{z=-\hat{\rho}d}\mathrm{d}x + \int_{\ell}^{L}\left(\frac{\partial \phi^{(p)}}{\partial z}\overline{\varphi^{(h)}} - \frac{\partial \overline{\varphi^{(h)}}}{\partial z}\phi^{(p)}\right)\Bigg|_{z=-\hat{\rho}d}\mathrm{d}x \\ &= \int_0^{\ell} W^{(p)}(x)\overline{\varphi^{(h)}}(x, -\hat{\rho}d)\,\mathrm{d}x - \int_{\ell}^{L}\frac{x - x^c}{L-\ell}\overline{\varphi^{(h)}}(x, -\hat{\rho}d) + \phi^{(p)}(x, -\hat{\rho}d)\,\mathrm{d}x. \end{aligned} \tag{S.4.2}$$

Therefore, we can write

$$F^{(p,h)} = -\frac{\ell^2}{d(L-\ell)}A^{(p,h)} + C^{(p,h)} \tag{S.4.3}$$

where

$$A^{(p,h)} = \frac{1}{\ell}\int_0^{\ell} W^{(p)}(x)\overline{G^{(h)}(x)}\,\mathrm{d}x \tag{S.4.4}$$

with $G^{(h)}$ defined by (3.14) and

$$C^{(p,h)} = \frac{\ell}{d(L-\ell)}\int_{\ell}^{L}\frac{x - x^c}{L-\ell}\overline{G^{(h)}(x)}\,\mathrm{d}x. \tag{S.4.5}$$

Using the definition (3.14) with (S.3.10) allows us to write

$$C^{(p,h)} = \frac{\ell^2}{d(L-\ell)}\sum_{m=-\infty}^{\infty}\frac{\overline{g_m{}^{(h)}}g_m{}^{(p)}}{\beta_m L}. \tag{S.4.6}$$

In Appendix A we show that

$$C^{(p,h)} = -\mathrm{i}\frac{(L-\ell)}{12d}. \tag{S.4.7}$$

We now consider the reciprocal heave-induced pitch force. From (2.13) we express the heave-induced pitch force as

$$F^{(h,p)} = -\frac{1}{d(L-\ell)^2}\int_\ell^L (x-x^c)\phi^{(h)}(x,-\hat{\rho}d)\,\mathrm{d}x + \frac{1}{d(L-\ell)^2}\int_{-\hat{\rho}d}^0 (z-z^c)\left(\phi^{(h)}(\ell,z) - \mathrm{e}^{\mathrm{i}kL}\phi^{(h)}(0,z)\right)\,\mathrm{d}z \tag{S.4.8}$$

after using the conditions satisfied by the $\phi^{(p)}$ on $z=-\hat{\rho}d$, $x=0,\ell$. Using $\phi^{(h)}$ and $\varphi^{(+p)}$ in Green's identity over the region V^+ we find

$$\begin{aligned}
0 &= \int_0^\ell \left(\frac{\partial\phi^{(h)}}{\partial z}\overline{\varphi^{(+p)}} - \frac{\partial\overline{\varphi^{(+p)}}}{\partial z}\phi^{(h)}\right)\Bigg|_{z=-\hat{\rho}d}\mathrm{d}x - \int_0^\ell \left(\frac{\partial\phi^{(h)}}{\partial z}\overline{\varphi^{(+p)}} - \frac{\partial\overline{\varphi^{(+p)}}}{\partial z}\phi^{(h)}\right)\Bigg|_{z=0}\mathrm{d}x \\
&\quad + \int_{-\hat{\rho}d}^0 \left(\frac{\partial\phi^{(h)}}{\partial x}\overline{\varphi^{(+p)}} - \frac{\partial\overline{\varphi^{(+p)}}}{\partial x}\phi^{(h)}\right)\Bigg|_{x=\ell}\mathrm{d}z - \int_{-\hat{\rho}d}^0 \left(\frac{\partial\phi^{(h)}}{\partial x}\overline{\varphi^{(+p)}} - \frac{\partial\overline{\varphi^{(+p)}}}{\partial x}\phi^{(h)}\right)\Bigg|_{x=0}\mathrm{d}z \\
&= -\int_0^\ell W^{(h)}(x)\overline{\varphi^{(+p)}}(x,-\hat{\rho}d)\,\mathrm{d}x - \int_{-\hat{\rho}d}^0 \frac{(z-z^c)}{(L-\ell)}\left(\phi^{(h)}(\ell,z) - \mathrm{e}^{\mathrm{i}kL}\phi^{(h)}(0,z)\right)\,\mathrm{d}z.
\end{aligned} \tag{S.4.9}$$

Also, using $\phi^{(h)}$ and $\varphi^{(-p)}$ in Green's identity in the domain V^- gives

$$\begin{aligned}
0 &= \int_0^L \left(\frac{\partial\phi^{(h)}}{\partial z}\overline{\varphi^{(p)}} - \frac{\partial\overline{\varphi^{(p)}}}{\partial z}\phi^{(h)}\right)\Bigg|_{z=-\hat{\rho}d}\mathrm{d}x \\
&= \int_0^\ell W^{(h)}(x)\overline{\varphi^{(p)}}(x,-\hat{\rho}d)\,\mathrm{d}x + \int_\ell^L \overline{\varphi^{(-p)}}(x,-\hat{\rho}d) + \frac{(x-x^c)}{(L-\ell)}\phi^{(h)}(x,-\hat{\rho}d)\,\mathrm{d}x.
\end{aligned} \tag{S.4.10}$$

Using both (S.4.9) and (S.4.10) in (S.4.8) along with (S.3.13) results in

$$F^{(h,p)} = -\frac{\ell^2}{d(L-\ell)}A^{(h,p)} + C^{(h,p)} \tag{S.4.11}$$

where we have defined

$$A^{(h,p)} = \frac{1}{\ell}\int_0^\ell W^{(h)}(x)\overline{G^{(p)}(x)}\,\mathrm{d}x \tag{S.4.12}$$

and

$$C^{(h,p)} = \frac{1}{d(L-\ell)}\int_\ell^L \overline{\varphi^{(-p)}}(x,-\hat{\rho}d)\,\mathrm{d}x = \frac{\ell^2}{d(L-\ell)}\sum_{m=-\infty}^{\infty}\frac{\overline{g_m^{(p)}}g_m^{(h)}}{\beta_m L} \tag{S.4.13}$$

which follows from use of (3.8) and (S.3.9). Clearly $C^{(h,p)} = \overline{C^{(p,h)}}$ whilst $A^{(h,p)} = \overline{A^{(p,h)}}$ using the same reasoning outlined in the surge/heave section.

4.1 Numerical approximation

We simply have

$$\tilde{A}^{(h,p)} \approx \sum_{n=0}^{N} A_n^{(h)}\overline{E_n^{(p)}} \tag{S.4.14}$$

and

$$\tilde{A}^{(p,h)} \approx \sum_{n=0}^{N} A_n^{(p)}\overline{E_n^{(h)}} \tag{S.4.15}$$

where $A_n^{(h,p)}$ are the coefficients determined from (3.23) and (S.3.26) and $E_n^{(h,p)}$ are defined by (3.29) and (S.3.27) respectively.

4.2 Asymptotic results for small gaps

Using the leading order terms in the integral equations, we find

$$A^{(h,p)} \approx \frac{1}{\epsilon^2}\frac{\overline{\tilde{G}^{(p)}}\tilde{G}^{(h)}}{\tilde{T}}. \tag{S.4.16}$$

Using (3.35), (3.37) and (S.3.31) and simplifying results whilst using $C^{(h,p)} = \mathrm{i}L/12d$ from (S.4.7), we have from (S.4.11),

$$F^{(h,p)} = i\left(\frac{L}{12d} + \frac{\hat{\rho}d}{2L}\frac{\frac{\hat{\rho}^2}{3} - \frac{\hat{\rho}}{2} + \frac{1-\hat{\rho}}{Kd}}{\frac{K\hat{\rho}d-1}{Kd} + \frac{\epsilon}{2}\cot\left(\frac{kL}{2}\right)}\right) \tag{S.4.17}$$

is the leading order heave-induced pitch force for small gaps. $F^{(p,h)}$ is the conjugate of (S.4.17)

5 Combined surge-pitch forces

Following the methodology used for the previous problems we are able to show that

$$F^{(s,p)} = -\frac{\ell^2}{d(L-\ell)}A^{(s,p)} + C^{(s,p)} \tag{S.5.1}$$

where

$$A^{(s,p)} = \frac{1}{\ell}\int_0^\ell W^{(s)}(x)\overline{G^{(p)}}(x)\,\mathrm{d}x \tag{S.5.2}$$

and

$$C^{(s,p)} = \frac{1}{d(L-\ell)}\int_{-\hat{\rho}d}^0 \overline{\varphi^{(+p)}}(\ell,z) - \mathrm{e}^{-\mathrm{i}kL}\overline{\varphi^{(+p)}}(0,z)\,\mathrm{d}z, \tag{S.5.3}$$

Likewise, for the reciprocal force, we have

$$F^{(s,p)} = -\frac{\ell^2}{d(L-\ell)}A^{(s,p)} + C^{(s,p)} \tag{S.5.4}$$

where

$$A^{(s,p)} = \frac{1}{\ell}\int_0^\ell W^{(p)}(x)\overline{G^{(s)}}(x)\,\mathrm{d}x \tag{S.5.5}$$

and

$$C^{(p,s)} = \frac{1}{d(L-\ell)^2}\int_{-\hat{\rho}d}^0 (z-z^c)\left(\overline{\varphi^{(s)}}(\ell,z) - \mathrm{e}^{-\mathrm{i}kL}\overline{\varphi^{(s)}}(0,z)\right)\,\mathrm{d}z. \tag{S.5.6}$$

Here we deduce from (S.5.6), after lengthy algebra,

$$\begin{aligned} C^{(s,p)} \quad = \quad & -\frac{\ell d\hat{\rho}(1-\hat{\rho})}{2(L-\ell)^2} + \frac{2\hat{\rho}d^3\sin^2(kL/2)}{\ell(L-\ell)^2}\left(\frac{\hat{\rho}^3}{4} - \frac{\hat{\rho}^2}{3} + (1-\hat{\rho})\left(\frac{\hat{\rho}}{Kd} + \frac{\ell^2}{6d^2}\right)\right) \\ & + \frac{4}{d\ell(L-\ell)^2}\sum_{m=1}^{\infty}\frac{(1-(-1)^m\cos kL)(1-\cosh(\alpha_m\hat{\rho}d) + (\frac{1}{2}-\hat{\rho})\alpha_m d\sinh(\alpha_m\hat{\rho}d))}{\alpha_m^4((\alpha_m/K)\sinh(\alpha_m\hat{\rho}d) - \cosh(\alpha_m\hat{\rho}d))}. \end{aligned} \tag{S.5.7}$$

We remark that a calculation of $C^{(s,p)}$ reveals it to be identical to $C^{(p,s)}$, which is expected by reciprocity.

5.1 Numerical approximation

We have

$$A^{(s,p)} \approx \sum_{n=0}^{N} A_n^{(s)}\overline{E_n^{(p)}} \tag{S.5.8}$$

and

$$A^{(p,s)} \approx \sum_{n=0}^{N} A_n^{(p)}\overline{E_n^{(s)}} \tag{S.5.9}$$

and, of course, $A^{(p,s)} = \overline{A^{(s,p)}}$, for reasons already detailed.

5.2 Asymptotic results for small gaps

First we note from (S.5.7) that

$$C^{(p,s)} \approx \frac{1}{\epsilon}\frac{2\hat{\rho}^2 d^2 \sin^2(kL/2)}{L^2}\left(\frac{\hat{\rho}^2}{4} - \frac{\hat{\rho}}{3} + \frac{(1-\hat{\rho})}{Kd}\right) \tag{S.5.10}$$

The leading order approximation to the solution of the integral equation is used to find

$$A^{(p,s)} \approx \frac{1}{\epsilon^3}\frac{\tilde{G}^{(p)}\overline{\tilde{G}^{(s)}}}{\tilde{T}}. \tag{S.5.11}$$

Using (S.1.22), (S.3.31) and (3.35) and simplifying, we can combine our result with (S.5.10) in (S.5.4) to get

$$F^{(s,p)} = \frac{1}{\epsilon}\frac{\hat{\rho}^2 d}{KL^2}\left(2Kd\left(\frac{\hat{\rho}^2}{4} - \frac{\hat{\rho}}{3} + \frac{1-\hat{\rho}}{Kd}\right) - \frac{K\hat{\rho}d - 2}{\frac{K\hat{\rho}d-1}{Kd} + \frac{1}{2}\epsilon\cot\left(\frac{kL}{2}\right)}\left(\frac{\hat{\rho}^2}{3} - \frac{\hat{\rho}}{2} + \frac{1-\hat{\rho}}{Kd}\right)\right)\sin^2\left(\frac{kL}{2}\right). \tag{S.5.12}$$

Appendix A: Evaluation of infinite series

In this Appendix we establish three main non-trivial results using the following definitions given in the body of the paper and the supplementary material: $\beta_m = k + 2m\pi/L$,

$$g_m^{(h)} = \mathrm{i}(\mathrm{e}^{-\mathrm{i}\beta_m L} - \mathrm{e}^{-\mathrm{i}\beta_m \ell})/(\beta_m \ell) \tag{S.A.1}$$

and

$$g_m^{(p)} = \frac{\mathrm{i}}{2\beta_m \ell}(\mathrm{e}^{-\mathrm{i}\beta_m L} + \mathrm{e}^{-\mathrm{i}\beta_m \ell}) + \frac{1}{\beta_m^2 \ell a}(\mathrm{e}^{-\mathrm{i}\beta_m L} - \mathrm{e}^{-\mathrm{i}\beta_m \ell}). \tag{S.A.2}$$

where we have used the abbreviation $a = L - \ell$. Then

$$\text{(i) } C^{(h,h)} = \frac{\ell^2}{da}\sum_{m=-\infty}^{\infty}\frac{\left|g_m^{(h)}\right|^2}{\beta_m L} = \frac{a\cot(kL/2)}{2d} \tag{S.A.3}$$

is the key result from which it will be shown that

$$\text{(ii) } C^{(p,p)} = \sum_{m=-\infty}^{\infty}\frac{\left|g_m^{(p)}\right|^2}{\beta_m L} = 0 \tag{S.A.4}$$

and

$$\text{(iii) } C^{(p,h)} = \frac{\ell^2}{da}\sum_{m=-\infty}^{\infty}\frac{g_m^{(p)}\overline{g_m^{(h)}}}{\beta_m L} = \mathrm{i}\frac{a^2}{12\ell^2}. \tag{S.A.5}$$

First, from (S.A.3) we note that

$$C^{(h,h)} = \frac{L^2}{da}J_1(a), \qquad \text{where } J_1(a) = \sum_{m=-\infty}^{\infty}\frac{\sin^2(\beta_m a/2)}{(\beta_m L)^3}. \tag{S.A.6}$$

Then clearly $J(0) = 0$ and

$$J_1'(a) = \frac{2}{L}\sum_{m=-\infty}^{\infty}\frac{\sin(\beta_m a)}{(\beta_m L)^2}, \tag{S.A.7}$$

so that $J_1'(0) = 0$ and then finally

$$J_1''(a) = \frac{2}{L^2}\sum_{m=-\infty}^{\infty}\frac{\cos(\beta_m a)}{\beta_m L}, \tag{S.A.8}$$

which we choose to write as

$$J_1''(a) = \frac{2a}{L^3}\sum_{m=-\infty}^{\infty}\frac{\cos(ka + \lambda m)}{ka + \lambda m}, \tag{S.A.9}$$

where we let $\lambda = 2\pi a/L$ so $0 < \lambda < 2\pi$. Now consider the function

$$F(z) = \frac{\pi \cos(ka + (\lambda - \pi)z)}{(ka + \lambda z)\sin(\pi z)}. \tag{S.A.10}$$

Clearly $F(z)$ has poles at $z = n \in \mathbb{Z}$ and at $z = -ka/\lambda = -kL/2\pi$ where we have assumed $kL \neq 2\pi n$ so all poles are simple poles. If we let $z = R\mathrm{e}^{\mathrm{i}\theta}$, it is clear that as $|z| \to \infty$ (i.e. $R \to \infty$) that we have

$$|F(z)| \sim \frac{\pi \mathrm{e}^{|\lambda - \pi||z|}}{\lambda |z| \mathrm{e}^{\pi |z|}}, \tag{S.A.11}$$

which is clearly exponentially decaying and thus we can write

$$0 = \lim_{N\to\infty} \oint_{C_N} F(z)\, \mathrm{d}z \tag{S.A.12}$$

where C_N is a circle of radius $N + \frac{1}{2}$ centred on the origin of the complex plane. Computing the residues inside C_N and letting $N \to \infty$ results in the identity

$$\sum_{m=-\infty}^{\infty} \frac{\cos(ka + \lambda m)}{ka + \lambda m} = \frac{L}{2a} \cot\left(\frac{kL}{2}\right). \tag{S.A.13}$$

Thus, we have established that

$$J_1''(a) = \frac{1}{L^2} \cot\left(\frac{kL}{2}\right). \tag{S.A.14}$$

Integrating twice with respect to a and applying the conditions $J_1'(0) = J_1(0) = 0$ gives

$$J_1(a) = \frac{a^2}{2L^2} \cot\left(\frac{kL}{2}\right) \tag{S.A.15}$$

and (i) follows.

In order to calculate result (ii) start with the definition

$$J_2(a) = \sum_{m=-\infty}^{\infty} \frac{\cos^2(\beta_m a/2)}{(\beta_m L)^3} \tag{S.A.16}$$

and so we can write

$$J_2(a) = \sum_{m=-\infty}^{\infty} \frac{1}{(\beta_m L)^3} - \sum_{m=-\infty}^{\infty} \frac{\sin^2(\beta_m a/2)}{(\beta_m L)^3} = \frac{1}{8} \cot\left(\frac{kL}{2}\right) \operatorname{cosec}^2\left(\frac{kL}{2}\right) - \frac{1}{4} J_1(a), \tag{S.A.17}$$

using Gradshteyn and Ryzhik (1981). Additionally, let us define

$$J_3(a) = \sum_{m=-\infty}^{\infty} \frac{\sin(\beta_m a/2)\cos(\beta_m a/2)}{(\beta_m L)^4} \tag{S.A.18}$$

so that

$$J_3'(a) = \frac{1}{2L} \sum_{m=-\infty}^{\infty} \frac{1}{(\beta_m L)^3} - \frac{1}{4L} J_1(a) = \frac{1}{16L} \cot\left(\frac{kL}{2}\right) \left(\operatorname{cosec}^2\left(\frac{kL}{2}\right) - \frac{2a^2}{L^2}\right). \tag{S.A.19}$$

Integrating with respect to a and applying $J_3(a) = 0$ gives

$$J_3(a) = \frac{a}{16L} \cot\left(\frac{kL}{2}\right) \left(\operatorname{cosec}^2\left(\frac{kL}{2}\right) - \frac{2}{3}\frac{a^2}{L^2}\right). \tag{S.A.20}$$

Finally, letting

$$J_4(a) = \sum_{m=-\infty}^{\infty} \frac{\sin^2(\beta_m a/2)}{(\beta_m L)^5} \tag{S.A.21}$$

we see that $J_4'(a) = J_3(a)/L$ and $J_4(a) = 0$ so that we can deduce

$$J_4(a) = \frac{a^2}{32L} \cot\left(\frac{kL}{2}\right) \left(\operatorname{cosec}^2\left(\frac{kL}{2}\right) - \frac{1}{3}\frac{a^2}{L^2}\right). \tag{S.A.22}$$

From the definition (S.A.4) we find that

$$C^{(p,p)} = \sum_{m=-\infty}^{\infty} \frac{\left|g_m{}^{(p)}\right|^2}{\beta_m L} = \frac{aL^2}{\ell^2} J_2(a) - \frac{4L^3}{\ell^2} J_3(a) + \frac{4L^4}{\ell^2 a} J_4(a) \tag{S.A.23}$$

and (iii) follows.

Moving on, we define

$$J_5(a) = \sum_{m=-\infty}^{\infty} \frac{\cos(\beta_m a)}{(\beta_m L)^4} \tag{S.A.24}$$

so that $J_5(0) = 0$ and

$$J_5'(a) = \frac{1}{L} \sum_{m=-\infty}^{\infty} \frac{\cos(\beta_m a)}{(\beta_m L)^2} \tag{S.A.25}$$

so that $J_5'(0) = (1/4L)\text{cosec}^2(kL/2)$. Finally,

$$J_5''(a) = \frac{1}{L^2} \sum_{m=-\infty}^{\infty} \frac{\sin(\beta_m a)}{\beta_m L} \tag{S.A.26}$$

or

$$J_5''(a) = -\frac{a}{L^3} \sum_{m=-\infty}^{\infty} \frac{\sin(ka + \lambda m)}{ka + \lambda m} \tag{S.A.27}$$

where we have let $\lambda = 2\pi a/L$ so $0 < \lambda < 2\pi$. Similarly to before we define

$$G(z) = \frac{\pi \sin(ka + (\lambda - \pi)z)}{(ka + \lambda z)\sin(\pi z)} \tag{S.A.28}$$

so that $G(z)$ has poles at $z = n \in \mathbb{Z}$ and at $z = -ka/\lambda = -kL/2\pi$ and we assume $kL \neq 2\pi n$ so all poles are simple poles. Again, it can be shown that $|G(z)| \to 0$ as $|z| \to \infty$ and so

$$0 = \lim_{N\to\infty} \oint_{C_N} G(z)\, \mathrm{d}z \tag{S.A.29}$$

where C_N is a circle of radius $N + \frac{1}{2}$. Calculating the residues from the poles inside C_N and letting $N \to \infty$ results in

$$\sum_{m=-\infty}^{\infty} \frac{\sin(ka + \lambda m)}{ka + \lambda m} = \frac{\pi}{\lambda} = \frac{L}{2a}. \tag{S.A.30}$$

and hence

$$J_5''(a) = -\frac{1}{2L^2}. \tag{S.A.31}$$

Integrating twice with respect to a and applying our the initial conditions gives

$$J_5(a) = -\frac{a}{4L}\left(\frac{a}{L} + \text{cosec}^2\left(\frac{kL}{2}\right)\right). \tag{S.A.32}$$

Finally consider

$$J_6(a) = \sum_{m=-\infty}^{\infty} \frac{\sin(\beta_m a)}{(\beta_m L)^3} \tag{S.A.33}$$

from which it follows that

$$J_6'(a) = -\frac{1}{L} \sum_{m=-\infty}^{\infty} \frac{\sin(\beta_m a)}{\beta_m L} = -\frac{1}{L} R(a) = \frac{a}{4L^2}\left(\frac{a}{L} + \text{cosec}^2\left(\frac{kL}{2}\right)\right) \tag{S.A.34}$$

and so

$$J_6(a) = \frac{a^2}{24L^2}\left(\frac{2a}{L} + 3\,\text{cosec}^2\left(\frac{kL}{2}\right)\right) + J_6(0), \tag{S.A.35}$$

$$C^{(h,p)} = \frac{\mathrm{i}L^2}{da}\left(P_5(a) + 2\frac{L}{a}\left(P_6(a) - P_6(0)\right)\right). \tag{S.A.36}$$

and the result in (iii) follows.

Appendix B: Another infinite series

We start with

$$\sum_{m=-\infty}^{\infty} \frac{e^{i\beta_m(x-x')}}{\beta_m L} = \sum_{m=-\infty}^{\infty} \frac{\cos(kX+\hat{\lambda}m)}{kX+\hat{\lambda}m} + i \sum_{m=-\infty}^{\infty} \frac{\sin(kX+\hat{\lambda}m)}{kX+\hat{\lambda}m}, \tag{S.B.1}$$

where we have let $X = (x - x') > 0$ (without loss of generality) and $\hat{\lambda} = 2\pi X/L$ such that $0 < X \leq \ell$ and $0 < \hat{\lambda} < 2\pi$. Using (S.A.13) and (S.A.30) from Appendix A we have

$$\sum_{m=-\infty}^{\infty} \frac{e^{i\beta_m X}}{\beta_m L} = \frac{1}{2}\left(\cot\left(\frac{kL}{2}\right) + i\right). \tag{S.B.2}$$

Conjugating this result is the same as reversing the sign of X and if $X = 0$ the series is real. Then the general result is

$$\sum_{m=-\infty}^{\infty} \frac{e^{i\beta_m(x-x')}}{\beta_m L} = \frac{1}{2}\cot\left(\frac{kL}{2}\right) + \frac{i}{2}\,\mathrm{sgn}(x-x'). \tag{S.B.3}$$

where $\mathrm{sgn}(X) = -1, 0, 1$ when $X < 0, X = 0, X > 0$ respectively.